

(2½ Hours)

[Total Marks: 75]

- N. B.: (1) **All** questions are **compulsory**.  
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.  
 (3) Answers to the **same question** must be **written together**.  
 (4) Numbers to the **right** indicate **marks**.  
 (5) Draw **neat labeled diagrams** wherever **necessary**.  
 (6) Use of **Non-programmable** calculators is **allowed**.

**1. Attempt any three of the following:****15**

- Explain 3D's (Defense, Detection, and Deterrence) aspects of security can be applied to any situation.
- Explain various Application-layer attacks which include any exploit directed at the applications running on top of the OSI protocol stack.
- Write a short note on CIA Triad Model with reference to Security in Computing.
- With the help of diagram, explain how Onion Defence Model is better than other Model for security.
- What is meant by Zone of Trust? Explain the importance of Zone of Trust for communication through with diagram.
- What are the various countermeasures that, anyone can implement to minimize the risk of a successful attack?

**2. Attempt any three of the following:****15**

- Explain different types of Authentication in detail.
- How Kerberos Authentication Process takes place? Explain each step with diagram.
- Write a short note on Certificate-Based Authentication.
- What is meant by Extensible Authentication Protocol (EAP)? Explain its different types.
- Explain role of PKI (Public Key Infrastructure) in Security in Computing and Structure and Function of PKI.
- "Each layer of security is designed for a specific purpose and can be used to provide authorization rules". Explain this statement with reference to Database Security Layers and its types.

**3. Attempt any three of the following:****15**

- Explain different layers of two-tier network fundamentals.
- With reference to OSI model in which layer does Router operate? Explain the working of Routing Protocols.
- Write a short note on different generation of Firewalls.
- Explain role of ICMP, SNMP and ECHO in network hardening.
- With the help of diagram, explain working of Bluetooth Protocol Stack.
- What is meant by Wireless Intrusion Detection and Prevention? Explain working of it.

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4. Attempt any three of the following: 15
- a Write a short note on two types of IDS Generation in brief.
  - b What is Private Branch Exchange (PBX)? Explain how it can be secured
  - c How Mandatory Access Control Lists (MACL) differ from Discretionary access control lists (DACLS)? Explain.
  - d Explain working of Biba and Clark Wilson Classic Security Models.
  - e What is meant by Security Reference Monitor? Explain Windows Security Reference Monitor in detail.
  - f Explain main problems of TCP/IP's lack of security.
5. Attempt any three of the following: 15
- a. What is meant by Hypervisor machine? Explain Why it is necessary to protect this machine.
  - b. Write a short note on Security Benefits of Cloud Computing.
  - c. With the help of diagram explain the concept of Secure development lifecycle in Agile.
  - d. Explain phishing mechanism and 3D's aspects of security with reference to it.
  - e. Give a reason in brief, why it is mandatory to update application patches? Explain various mechanisms for easily updating applications.
  - f. Explain various concerns for web application security to be considered with reference to Security in Computing.